

$$T3. (1) E(X_k) = 0 \quad D(X_k) = E(X_k^2) = \int_{-0.5}^{0.5} x_k^2 p_k dx_k$$

$$= \int_{-0.5}^{0.5} x^2 dx = \left(\frac{1}{3} x^3 \right) \Big|_{-0.5}^{0.5}$$

$$= \frac{1}{3} \left(\frac{1}{8} + \frac{1}{8} \right) = \frac{1}{12}$$

由中心极限定理, 当 n 无限增大时, 独立同分布总和为正态分布

$$Z = \frac{\sum_{k=1}^{1500} X_k - 0}{\sqrt{\frac{1}{12} \cdot 1500}} = \frac{\sum_{k=1}^{1500} X_k}{\frac{10\sqrt{5}}{2}} = \frac{\sum_{k=1}^{1500} X_k}{5\sqrt{5}} \quad \text{近似服从正态分布 } N(0, 1)$$

$$\text{记 } X = \sum_{k=1}^{1500} X_k$$

$$P(X > 15 \text{ 或 } X < -15) \Rightarrow Z = \frac{15-0}{5\sqrt{5}} = \frac{3}{\sqrt{5}} = 1.34$$

$$= [1 - \Phi(1.34)] \times 2 = (1 - 0.9099) \times 2 = 0.0901 \times 2 = 0.1802$$

$$(2) Z = \frac{10-0}{5\sqrt{\frac{1}{12}} \sqrt{n}} = \frac{10}{\sqrt{\frac{1}{12}} \sqrt{n}} \quad n \text{ 最多为 } 443$$

$$P(-10 < X < 10) = 1 - [1 - \Phi(\frac{10}{\sqrt{\frac{1}{12}} \sqrt{n}})] \times 2 \geq 0.9$$

$$0.1 \geq [1 - \Phi(\frac{10}{\sqrt{\frac{1}{12}} \sqrt{n}})] \times 2$$

$$\frac{200}{33}$$

$$0.05 \geq 1 - \Phi(\frac{10}{\sqrt{\frac{1}{12}} \sqrt{n}}) \quad \Phi(\frac{10}{\sqrt{\frac{1}{12}} \sqrt{n}}) \geq 0.95$$

$$\text{则 } \frac{10}{\sqrt{\frac{1}{12}} \sqrt{n}} \geq 1.65 \quad \frac{1000}{165} \geq \sqrt{\frac{n}{12}} \quad 20.99 \geq \sqrt{n} \quad 440.77 \geq n$$

n 最多为 440 个数相加

$$X \sim b(1, 0.8)$$

T5 记从其中取出 X 根长度不小于 3m. 为则有 ~~$X \sim b(1, 0.8)$~~
满足 ~~重~~ 伯努利试验.

则 $X=0$ 为长度小于 3m. $X=1$ 为长度大于 3m.

则其中至少有 30 根短于 3m 的概率 $P(A) = P(\sum_{i=1}^{100} X_i \leq 70)$

$$X_i \sim b(1, 0.8) \quad E(X_i) = 0.8 \quad D(X_i) = p(1-p) = 0.16$$

$$E(\sum_{i=1}^{100} X_i) = \sum_{i=1}^{100} E(X_i) = 80 \quad D(\sum_{i=1}^{100} X_i) = \sum_{i=1}^{100} D(X_i) = 16$$

$$\sum_{i=1}^{100} X_i \sim N(80, 16)$$

$$P(\sum_{i=1}^{100} X_i \leq 70) = P(\frac{\sum_{i=1}^{100} X_i - 80}{4} \leq \frac{70-80}{4}) = \Phi(-2.5) = 1 - \Phi(2.5)$$

$$= 1 - 0.9938 = 0.0062.$$



指数分布 参数 $\theta > 0$, 数学期望 θ , 方差 θ^2

6. $E(X+Y) = 0.2 + 0.3 = 0.5$

$$D(X+Y) = D(X) + D(Y) + 2E\{[X-E(X)][Y-E(Y)]\}$$

$\because X, Y$ 独立 $\therefore D(X+Y) = D(X) + D(Y) = 0.04 + 0.09 = 0.13$

对于独立同分布 共需时间为 $V = \sum_{k=1}^{20} V_k$ 单台机器修理所需时间 $V_k = X+Y, E(V_k) = 0.5, D(V_k) = 0.13$.

$$Z = \frac{\sum_{k=1}^{20} V_k - \sum_{k=1}^{20} E(V_k)}{\sqrt{0.13 \times 20}} \Rightarrow \frac{8 - 20 \times 0.5}{\sqrt{0.13 \times 20}} = \frac{-2}{\sqrt{2.6}} = -1.24$$

由中心极限定理

他在 8h 内完成的概率 $P(V_k \leq 8) = \Phi(-1.24) = 1 - \Phi(1.24)$
 $= 1 - 0.8925 = 0.1075$

$$\begin{array}{r} 1.0000 \\ 0.8925 \\ \hline 0.1075 \end{array}$$

8. 单个部件正常工作为两点分布. 有 Y 个部件正常工作 $Y \sim B(100, 0.9)$
 整个系统起作用 $Y \geq 85$.

$$E(Y) = np = 90, \sqrt{D(Y)} = \sqrt{np(1-p)} = \sqrt{100 \times 0.9 \times 0.1} = 3.$$

$$Z = \frac{Y - E(Y)}{\sqrt{D(Y)}} \quad Z_1 = \frac{85 - 90}{3} = \frac{-5}{3} \approx -1.6667$$

$$P(Y \geq 85) = 1 - \Phi(-1.6667) = \Phi(1.6667) \approx 0.9525.$$

记一周发生数为 $X_i, E(X_i) = 2.2, D(X_i) = 1.4^2$

9. $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{52} \sum_{i=1}^{52} X_i, E(\bar{X}) = \frac{1}{52} \sum_{i=1}^{52} E(X_i) = E(X_i)$

$$D(\bar{X}) = D\left(\frac{1}{52} \sum_{i=1}^{52} X_i\right) = \frac{1}{52^2} \sum_{i=1}^{52} D(X_i) = \frac{1.4^2}{52} = \frac{1.96}{52} \approx 0.0377$$

由中心极限定理.

$$\bar{X} \sim N(2.2, \frac{1.96}{52})$$

$$P(\bar{X} < 2) = P\left(\frac{\bar{X} - 2.2}{\sqrt{\frac{1.96}{52}}} < \frac{2 - 2.2}{\sqrt{\frac{1.96}{52}}}\right)$$

$$= \Phi\left(\frac{-0.2}{1.4 \sqrt{52}}\right) = \Phi\left(-\frac{\sqrt{52}}{7}\right) = 1 - \Phi\left(\frac{\sqrt{52}}{7}\right) \approx 1 - 0.8485 = 0.1515$$



(2) 一年事故发生数 $\sum_{i=1}^{52} X_i$: $E(\sum_{i=1}^{52} X_i) = 52 E(X_i) = 52 \times 0.2 = 10.4$

$$D(\sum_{i=1}^{52} X_i) = 52 D(X_i) = 52 \times 1.6^2$$

一年事故发生数小于100的概率: $\sum_{i=1}^{52} X_i \sim (10.4, 52 \times 1.6^2)$

$$P(\sum_{i=1}^{52} X_i < 100) = P(\frac{\sum_{i=1}^{52} X_i - 10.4}{\sqrt{52 \times 1.6}} < \frac{100 - 10.4}{\sqrt{52 \times 1.6}})$$

$$= \Phi(1.426) = \Phi(1.63) = 0.9236$$

$$= 0.9770$$

14. 记第 X_i 人治愈为 $X_i = 1$, 其无治愈为 $X_i = 0$.

则100个人治愈的人数为 $\sum_{i=1}^{100} X_i$

则 $X_i \sim b(1, p)$. $E(\sum_{i=1}^{100} X_i) = \sum_{i=1}^{100} E(X_i) = 100p$

$$D(\sum_{i=1}^{100} X_i) = \sum_{i=1}^{100} D(X_i) = 100p(1-p)$$

则由中心极限定理 $\sum_{i=1}^{100} X_i \sim N(100p, 100p(1-p))$

11. $p = 0.8$. $P(\sum_{i=1}^{100} X_i \geq 75) = 1 - P(\sum_{i=1}^{100} X_i \leq 75)$

$\sum_{i=1}^{100} X_i \sim N(80, 16)$

$$= 1 - P(\frac{\sum_{i=1}^{100} X_i - 80}{4} \leq \frac{75 - 80}{4})$$

接受断言概率为 0.8925

$$= 1 - \Phi(-1.25)$$

$$= 1 - (1 - \Phi(1.25)) = \Phi(1.25)$$

$$= 0.8944$$

$p = 0.7$

(2). $\sum_{i=1}^{100} X_i \sim N(70, 21)$ $P(\sum_{i=1}^{100} X_i \geq 75) = 1 - P(\sum_{i=1}^{100} X_i \leq 75)$

$$= \frac{75 - 70}{\sqrt{21}}$$

接受断言概率为 0.1379

$$= 1 - \Phi(1.097)$$

$$\frac{5}{\sqrt{21}}$$

$$= 1 - 0.8621$$

$$= 0.1379$$

